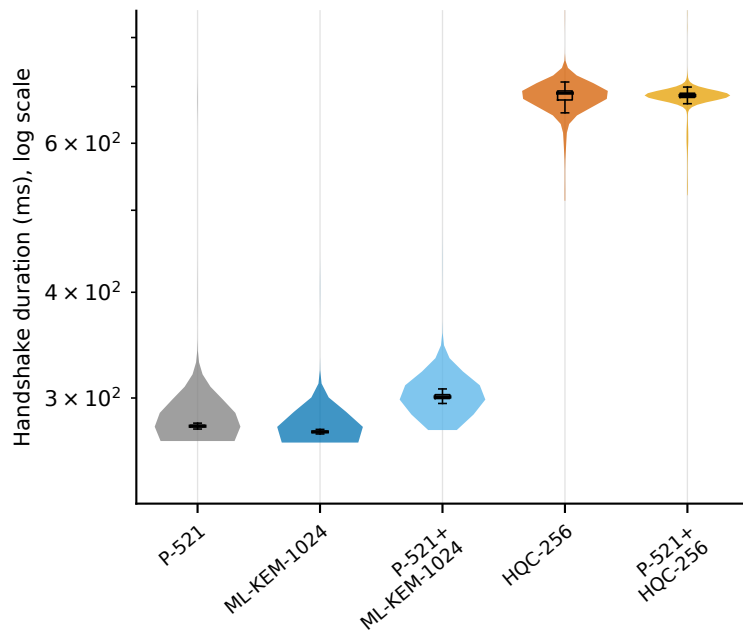
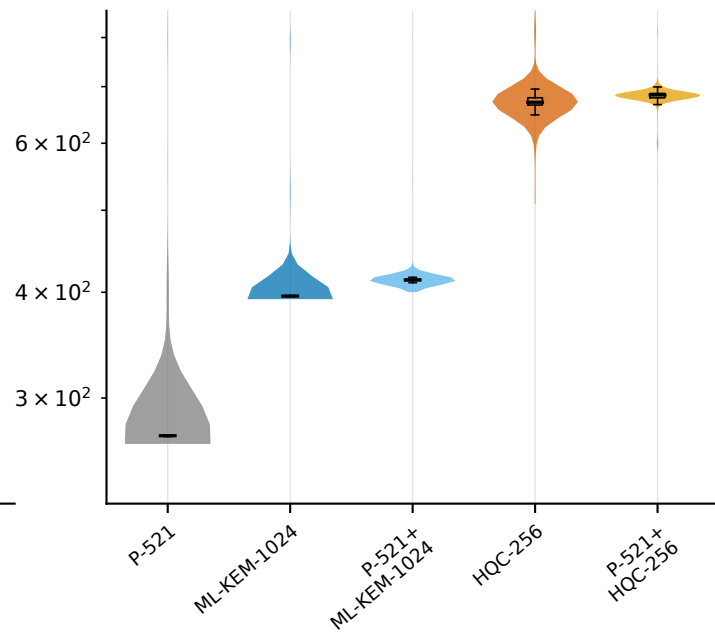


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	283.8	24.7	1.32k	285.4	40.8	1.69k
ML-KEM-1024	276.2	18.5	1.41k	401.6	16.4	1.68k
P-521+ML-KEM-1024	304.8	19.6	1.45k	414.8	4.7	809.0
HQC-256	686.9	10.3	1.99k	679.2	13.0	1.97k
P-521+HQC-256	680.8	4.4	944.2	682.9	2.8	822.3